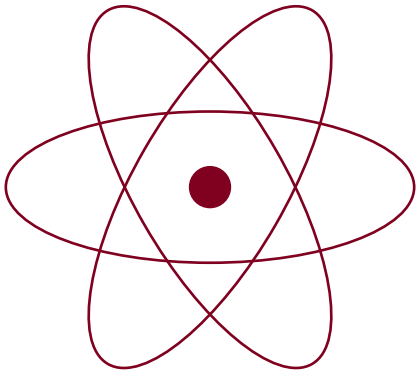
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Kinematics II

Advanced Olympiad Problem Set

Olympiad Training • Mechanics

SPhOH Team

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused and substantial collection of advanced problems for deliberate problem-solving practice in kinematics.

The handout is **problem-centered**. Explanations are intentionally brief and are included only where they help establish the ideas or techniques needed for the problems.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For difficult problems, several progressive hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

Difficulty Levels

The problems are organized strictly by difficulty level within each section:

- **Level I** — Fundamental problems involving direct algebraic calculations and introductory vector concepts.
- **Level II** — Multi-step problems combining moving frames, non-inertial systems, and geometric constraints.
- **Level III** — Challenging national problems requiring advanced analytical techniques.
- **Level IV** — Harder problems requiring more advanced mathematical knowledge.

Find more at: `https://devfhd.github.io/SPhOH/.`

CHAPTER 1

Kinematics II

Topic Note: Advanced Kinematics (Rolling, Relative Motion, and Constraints)
Advanced kinematics extends beyond basic trajectory equations to include reference frame transformations, rolling constraints, and circular kinematics:

- **Rolling Without Slipping:** For a cylinder or wheel of radius R rolling on a stationary surface, the velocity of the center of mass v_{cm} and angular velocity ω are related by $v_{\text{cm}} = \omega R$. The instantaneous center of rotation lies at contact points with zero local speed.
- **Relative Motion in Rotating and Accelerating Frames:** Galilean transformations for position, velocity, and acceleration allow analysis in non-inertial systems or moving constraints:
$$\vec{v}_{\text{absolute}} = \vec{v}_{\text{rel}} + \vec{V}_{\text{frame}}$$
- **Circular Kinematics:** Centripetal acceleration is given by $a_c = \frac{v^2}{r} = \omega^2 r$, and tangential acceleration by $a_t = \alpha r$.

1.1 Rolling Motion and Constraint Kinematics

PROBLEM 1

LEVEL I

Rotation Count of a Pure Rolling Drum

A uniform circular drum of radius $R = 0.50$ m rolls across a flat plane without slipping. If its central axis traverses a net distance of 31.4 m, how many full rotations did the drum complete? (Assume $\pi \approx 3.14$)

PROBLEM 2

LEVEL I

Linear Distance from Wheel Rotations

A circular wheel of diameter $D = 0.80$ m undergoes pure rolling along a horizontal road. If the

wheel turns through 450 complete revolutions, calculate the total horizontal distance traveled by the axle of the wheel.

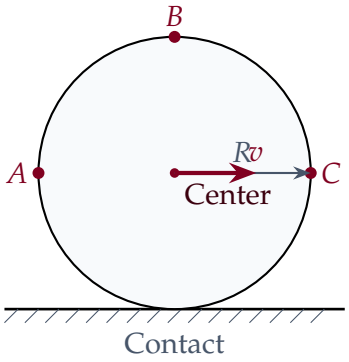
PROBLEM 3

LEVEL II

Velocities on a Rolling Rim

A solid cylinder with radius $R = 0.40$ m rolls smoothly across a horizontal surface without slipping. The center of mass advances at a constant speed of $v = 3.20$ m/s.

Determine the magnitude and direction of the velocity vectors for boundary points A (leftmost edge), B (topmost edge), and C (rightmost edge).

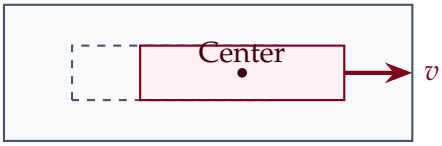


PROBLEM 4

LEVEL II

Kinematics of Unspooling Tape

A strip of adhesive tape lies flat on a table. One end is pulled horizontally backward at a constant speed $v = 14.0$ cm/s parallel to the surface. Determine the instantaneous speed of the geometric center of the active, moving segment of lifted tape.

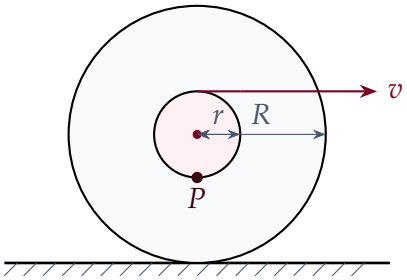


PROBLEM 5

LEVEL III

Kinematics of a Stepped Cylinder Spool

A stepped cylinder consists of a central axle of radius $r = R/3$ fitted inside two outer disks of radius R. A flexible thread is wound around the inner axle. The spool rests on a horizontal plane, and the free end of the thread is pulled horizontally to the right at a steady speed v. Calculate the exact linear velocity vector of the lowermost point P on the central axle under the no-slip condition.



PROBLEM 6

LEVEL III

Rolling Inside a Fixed Cylinder

A solid cylinder of radius r rolls without slipping along the inner surface of a fixed hollow cylinder of radius $R > r$. The center of the small cylinder moves with a constant angular speed Ω about the central axis of the outer cylinder.

(a) Find the angular velocity ω of the small cylinder about its own central axis.

(b) Determine the linear acceleration of the contact point on the small cylinder relative to a stationary observer.

1.2 Relative Motion and Moving Media

PROBLEM 7

LEVEL I

Relative Speed of Two Walkers

Two pedestrians walk along a straight path in the same direction. Pedestrian A walks at a steady pace of 4.0 1/h, while Pedestrian B walks ahead at 6.5 1/h. Calculate the velocity of Pedestrian B relative to Pedestrian A.

PROBLEM 8

LEVEL I

Catching Up on a Straight Highway

A highway patrol cruiser traveling west at 95.0 1/h pursues a speeder moving west at 80.0 1/h.

(a) What is the relative velocity of the speeder as observed from the patrol cruiser?

(b) What is the relative velocity of the patrol cruiser as observed from the speeder?

(c) If the vehicles are initially separated by 250 m, how long will it take for the patrol cruiser to catch the speeder?

PROBLEM 9

LEVEL I

Round-Trip Swimmer in a Current

A river flows with a uniform current of 0.500 m/s. A swimmer travels 1.00 upstream and returns to the starting location.

(a) If the swimmer maintains a speed of 1.20 m/s relative to still water, calculate the total time required for the round trip.

(b) How long would the same round trip take in perfectly still water?

(c) Explain physically why the presence of a river current always increases the total duration of a round trip.

PROBLEM 10

LEVEL I

Falling Object in an Accelerating Train

A bolt detaches from the ceiling of a train compartment that is accelerating northward along a straight track at 2.50 m/s^2.

(a) What is the acceleration vector of the bolt relative to the train?

(b) What is the acceleration vector of the bolt relative to the Earth?

(c) Describe the trajectory of the bolt as seen by a passenger seated in the train.

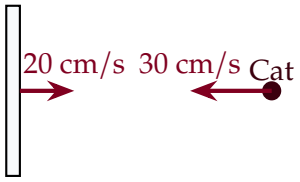
(d) Describe the trajectory of the bolt as seen by an observer standing on the ground.

PROBLEM 11

LEVEL I

Approach Rate to a Moving Plane Mirror

A cat walks directly toward a flat vertical mirror at 30 cm/s. Simultaneously, the mirror moves toward the cat at 20 cm/s. Find the rate at which the distance between the cat and its reflected image decreases.



PROBLEM 12

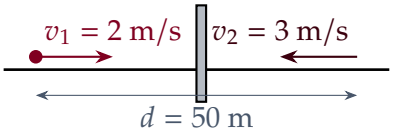
LEVEL II

Opposing Particles with Equal Acceleration

Two particles are initially separated by distance $d = 50$ m along a straight track. Particle 1 moves right with initial speed $v_1 = 2$ m/s, while Particle 2 moves left with initial speed $v_2 = 3$ m/s. Both particles experience a constant acceleration a directed to the left. If they collide after $t = 10$ s, find:

(a) The magnitude of acceleration a.

(b) The relative speed of the particles at the moment of impact.



PROBLEM 13

LEVEL II

Drifting Raft and Returning Motorboat

A motorboat traveling downstream along a river passes a floating log raft at point A. The boat continues downstream for $T = 60$ min, turns back, and travels upstream until it meets the raft

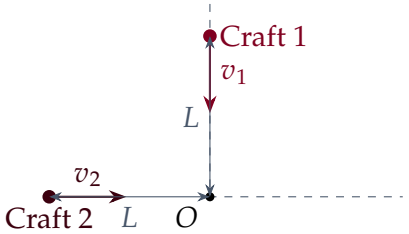
again at distance $L = 6.0$ downstream from A. Assuming the boat maintains constant speed relative to the water, calculate the speed of the river current.

PROBLEM 14

LEVEL II

Closest Approach of Perpendicular Craft

Two craft travel along perpendicular straight paths intersecting at point O. At a given instant, both are at distance $L = 20$ from O, approaching O with constant speeds v_1 and v_2. Find the minimum distance between them during their motion.



PROBLEM 15

LEVEL II

Optimal Path in a Wedge Enclosure

Two flat walls meet at an angle $2\alpha = 40°$. A runner starts at point P located along the bisector at distance $d = 10$ m from the corner. The runner must touch both walls in sequence before returning to P. If moving at $v = 2.0$ m/s, calculate the minimum time needed.

PROBLEM 16

LEVEL II

Cross-River Navigation and Minimum Drift

A river of width $W = 400$ m flows with a uniform current $v_r = 5.0$ m/s. A boat moves at speed $v_b = 3.0$ m/s relative to the water.

(a) Determine the heading angle θ relative to the bank that minimizes downstream drift.

(b) Calculate the minimum downstream drift distance experienced by the boat upon reaching the opposite bank.

PROBLEM 17

LEVEL III

Signal Timings on an Accelerating Train

A train of length $l = 420$ m accelerates from rest with $w = 2.5 \times 10^{-2}$ m/s^2. At $t_1 = 20$ s, a light at the front flashes (Event 1). At $t_2 = t_1 + \tau$ where $\tau = 45$ s, a light at the rear flashes (Event 2).

(a) Calculate the spatial distance between the two events in the train's frame.

(b) Calculate the distance between the two events in the ground frame.

(c) Find the constant velocity V of an inertial frame K where both events occur at identical spatial coordinates.

PROBLEM 18

LEVEL IV

Crawling Insect on a Reeled Elastic Thread

An elastic thread of resting length $L_0 = 1.20$ m has one end fixed to a wall. An insect crawls toward the wall at 6.0 mm/s relative to the thread material. At the same time, the outer end of the thread is reeled in at a constant rate $v_0 = 2.5$ cm/s. Calculate the maximum initial separation distance from the wall that still allows the insect to reach the wall.



PROBLEM 19

LEVEL IV

Relative Motion in Circular Paths

Two particles A and B move counterclockwise along concentric circular tracks of radii R_1 and R_2 ($R_2 > R_1$) with constant angular speeds ω_1 and ω_2, respectively. At $t = 0$, both particles lie on the positive x-axis. Find the rate of change of the distance $d(t)$ between them as a function of time, and determine the conditions under which $d(t)$ changes most rapidly.

1.3 Circular Motion and Rotating Systems

PROBLEM 20

LEVEL I

Centripetal Acceleration of a Tracked Vehicle

A model vehicle travels along a circular track of radius $r = 12.0$ m at a constant speed $v = 6.0$ m/s. Calculate the magnitude of its centripetal acceleration.

PROBLEM 21

LEVEL I

Angular Speed and Period of a Rotating Ring

A point on the outer rim of a rotating ring of radius $R = 0.50$ m experiences a constant centripetal acceleration of magnitude $a_c = 18.0\ \text{m/s}^2$.

(a) Compute the angular speed ω of the ring.

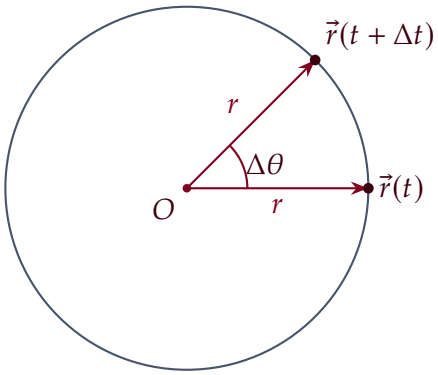
(b) Determine the period T of one complete rotation.

PROBLEM 22

LEVEL I

Geometric Derivation of Radial Acceleration

Derive the expression $a_c = v^2/r$ for uniform circular motion using vector geometry. Construct the velocity vector triangle corresponding to a small angular change $\Delta\theta$.



PROBLEM 23

LEVEL I

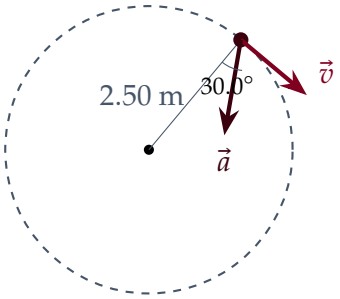
Accelerations in Non-Uniform Circular Motion

A particle moves clockwise along a circular track of radius $R = 2.50$ m. At a given instant, its total acceleration vector has magnitude $a = 15.0\ \text{m/s}^2$ and forms an angle of $30.0°$ with the inward radial direction.

(a) Calculate the radial acceleration component.

(b) Determine the instantaneous speed of the particle.

(c) Calculate the tangential acceleration component.



PROBLEM 24

LEVEL II

Accelerations on a Spinning Platform

Object A at radius $r_1 = 1.50$ m and object B at radius $r_2 = 2.50$ m rest on a turntable along the same radial line. At a given moment, object A has acceleration $\vec{a}_1 = (3.00\ \text{m/s}^2)\hat{i} + (4.00\ \text{m/s}^2)\hat{j}$. Determine the acceleration vector of object B at this moment.

PROBLEM 25

LEVEL II

Particle Slowing Down on a Curved Ramp

A particle enters a circular track of radius $R = 5.0$ m with an initial speed $v_0 = 10.0$ m/s. As it moves along the track, its tangential deceleration is constant and equal to $a_t = -2.0\ \text{m/s}^2$.

(a) Find the time t^* at which the radial and tangential accelerations have equal magnitude.

(b) Determine the distance traveled along the arc up to time t^*.

PROBLEM 26

LEVEL III

Reconstructing Circular Trajectories

A particle undergoes uniform circular motion in the xy-plane. At $t_1 = 3.00$ s, it is at $(4.00\text{ m}, 5.00\text{ m})$ with velocity $\vec{v}_1 = (2.50\text{ m/s})\hat{j}$ and acceleration pointing in the $+x$ direction. At $t_2 = 8.50$ s, its velocity is $\vec{v}_2 = (-2.50\text{ m/s})\hat{i}$ with acceleration pointing in the $+y$ direction. Assuming $t_2 - t_1$ is less than one period, find the center coordinates (x_c, y_c) of the orbit.

PROBLEM 27

LEVEL IV

Separation Rate of Clock Hands

The minute hand of a clock is three times longer than the hour hand. Find the time interval elapsed after 12:00 at which the distance between the tips of the hands increases at the maximum rate.

CHAPTER 2

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

Hint 1 | Problem 1 – Rotation Count of a Pure Rolling Drum

Relate the linear displacement of the center of mass to the angular displacement: $\Delta x = \theta R$, where $\theta = 2\pi N$ and N is the total number of revolutions.

Hint 2 | Problem 2 – Linear Distance from Wheel Rotations

The distance covered during pure rolling is equal to the total arc length swept along the perimeter: $x = N(\pi D)$.

Hint 3 | Problem 3 – Velocities on a Rolling Rim

Treat the bottom point of contact as the instantaneous center of zero velocity. Combine translational speed v with rotational velocity $\vec{\omega} \times \vec{r}$.

Hint 4 | Problem 4 – Kinematics of Unspooling Tape

Consider how the peeling boundary moves relative to the fixed surface and relate its rate of displacement to the pulled end.

Hint 5 | Problem 5 – Kinematics of a Stepped Cylinder Spool

Determine the angular velocity ω and center-of-mass velocity v_{cm} of the spool in terms of v using the instantaneous center of rotation at the contact floor.

Hint 6 | Problem 6 – Rolling Inside a Fixed Cylinder

Apply the rolling constraint at the contact boundary: the contact point on the inner cylinder has zero instantaneous tangential velocity relative to the outer fixed shell.

Hint 7 | Problem 7 – Relative Speed of Two Walkers

Use relative velocity in one dimension: $v_{B/A} = v_B - v_A$.

Hint 8 | Problem 8 – Catching Up on a Straight Highway

Define a coordinate system with west as positive. Express relative velocities as vector differences and compute time using $t = d/v_{\text{rel}}$.

Hint 9 | Problem 9 – Round-Trip Swimmer in a Current

The effective speed upstream is $v_{\text{rel}} - v_{\text{river}}$ and downstream is $v_{\text{rel}} + v_{\text{river}}$. Calculate $t = \frac{d}{v_{\text{up}}} + \frac{d}{v_{\text{down}}}$.

Hint 10 | Problem 10 – Falling Object in an Accelerating Train

In an accelerating frame, include pseudo-accelerations. Relative to Earth, only gravity acts on the free-falling bolt once detached.

Hint 11 | Problem 11 – Approach Rate to a Moving Plane Mirror

Work in the mirror's reference frame to determine the cat's relative velocity, then double this result to find the image closing speed.

Hint 12 | Problem 12 – Opposing Particles with Equal Acceleration

Since both particles experience the exact same acceleration vector, their relative acceleration is zero.

Hint 13 | Problem 13 – Drifting Raft and Returning Motorboat

Switch to a reference frame attached to the moving water. The raft remains at rest, and the boat takes equal time to move away and return.

Hint 14 | Problem 14 – Closest Approach of Perpendicular Craft

Express the squared separation $d(t)^2$ as a quadratic in time t and locate its minimum value.

Hint 15 | Problem 15 – Optimal Path in a Wedge Enclosure

Construct mirror reflections of point P across both walls to convert the triangular path into a straight line.

Hint 16 | Problem 16 – Cross-River Navigation and Minimum Drift

Since $v_b < v_r$, straight cross-river travel is impossible. Maximize the angle of the net velocity vector relative to the shore using geometric vector construction.

Hint 17 | Problem 17 – Signal Timings on an Accelerating Train

Compute ground positions for the front at t_1 and the rear at t_2, then apply Galilean transformations.

Hint 18 | Problem 18 – Crawling Insect on a Reeled Elastic Thread

Express position using the dimensionless variable $\xi(t) = x(t)/L(t)$ and integrate the differential equation $\dot{\xi}(t) = -u_{\text{rel}}/L(t)$.

Hint 19 | Problem 19 – Relative Motion in Circular Paths

Express positions vectorially as $\vec{r}_A(t)$ and $\vec{r}_B(t)$, write $d^2(t) = |\vec{r}_B - \vec{r}_A|^2$, and evaluate $\frac{\mathrm{d}}{\mathrm{d}t}[d(t)]$.

Hint 20 | Problem 20 – Centripetal Acceleration of a Tracked Vehicle

Apply the centripetal acceleration formula: $a_c = v^2/r$.

Hint 21 | Problem 21 – Angular Speed and Period of a Rotating Ring

Use $a_c = \omega^2 R$ to determine ω, then find period using $T = 2\pi/\omega$.

Hint 22 | Problem 22 – Geometric Derivation of Radial Acceleration

Show that the position triangle and velocity triangle for time interval Δt are similar isosceles triangles.

Hint 23 | Problem 23 – Accelerations in Non-Uniform Circular Motion

Decompose acceleration into $a_r = a\cos\theta$ and $a_t = a\sin\theta$, then relate radial acceleration to speed via $a_r = v^2/R$.

Hint 24 | Problem 24 – Accelerations on a Spinning Platform

Since both objects share the platform's rigid rotation rate ω, centripetal acceleration scales linearly with radius.

Hint 25 | Problem 25 – Particle Slowing Down on a Curved Ramp

Express speed as a function of time $v(t) = v_0 - |a_t|t$. Set radial acceleration $a_r = v(t)^2/R$ equal to $|a_t|$.

Hint 26 | Problem 26 – Reconstructing Circular Trajectories

Acceleration vectors point directly toward the center. Use velocity vectors to verify the turning sense of the path.

Hint 27 | Problem 27 – Separation Rate of Clock Hands

Construct parametric position vector functions for both hand tips, form $d^2(t)$, and optimize the rate of change $\frac{\mathrm{d}}{\mathrm{d}t}[d(t)]$.